Darsh Thakkar

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Professional Summary

Software and ML engineer with experience building AI systems, voice assistants, and full-stack applications. Skilled in Python, C++, React, and PyTorch with experience developing LLM systems, robotics interfaces, and privacy-focused AI tools.

Experience

Undergraduate Research Assistant – People and Robots Lab, UW-Madison *Feb 2026 – Present* | Madison, WI, USA

- Developed a user feedback canvas for a robot programming interface using Flask backend, enabling real-time message transmission and status visualization.
- Implemented status animations and blur effects for the programming workflow, including submit button functionality and retry state handling.
- Built React-based front-end components for robot programming interface with status panels, user satisfaction tracking, and condition switching capabilities.
- Programmed Temi robot for text display and scenario fallbacks using Docker and WSL environments for deployment.

AI and Software Engineer Intern — LexLegis AI *May 2025 - Jul 2025* | Mumbai, India

- Developed a fully offline AI legal assistant with chat, OCR, multilingual translation, and semantic search to ensure data privacy.
- Built cross-platform desktop features in C++, Python, and Qt/QML; integrated local LLM inference via llama.cpp and Llama Server.
- Implemented RAG architectures for grounded answer retrieval (RAG + BM25 + embeddings) across document collections.

Software Development Intern — Yodaplus *Jun 2024 - Aug 2024* | Mumbai, India

- Reviewed SQL queries and backend behavior to identify bugs and propose fixes; collaborated with engineers to resolve issues.
- Wrote Pandas-based data cleaning/validation scripts to improve data quality for downstream analytics and APIs.
- Developed full-stack features using Python, REST APIs, PostgreSQL, and React.

Projects

Voix — AI Voice Copilot for Windows (CalHacks 12.0)

- Built a voice operating system to control Windows apps, generate documents, and orchestrate workflows via natural conversation.
- Integrated Faster-Whisper (on-device STT), Groq + Claude (reasoning), and Python COM automation (Word/Excel/Chrome).
- Designed a floating desktop widget with asyncio, SoundDevice, and Silero VAD; added meeting recording, transcription, and action-item extraction locally.
- Achieved sub-second response latency while maintaining offline functionality and multi-app orchestration.

YouTube Gesture Remote (Computer Vision)

- Developed a real-time, offline gesture controller with hand pose recognition for playback control and an on-screen HUD.
- Implemented gesture set for play/pause, volume, full screen, and navigation; tuned gesture-to-action mapping for stability.

Education

University of Wisconsin-Madison — B.S. Computer Science and Data Science (Expected 2027)

GPA: 3.4 | Dean's List: Fall 2023, Fall 2024

Skills

Programming Languages: Python, C++, Java, JavaScript, SQL, R, C

ML/AI: PyTorch, scikit-learn, OpenCV, MediaPipe, NLP, CNNs, Reinforcement Learning

LLMs/RAG: llama.cpp, GGUF, embeddings, BM25, Tesseract OCR, Multilingual Translation, RAG architectures

Systems/Web: Qt/QML, ReactJS, REST APIs, PostgreSQL, SQLite, Git, CMake, asyncio, FFmpeg, PyInstaller, COM Automation, Flask, Docker, WSL

Certifications

Supervised Machine Learning (DeepLearning.AI/Stanford)